Core Measurements and Submission Timelines

The Executive Group of the Global Ocean Ship-based Hydrographic Investigations Program (GO-SHIP, www.go-ship.org) has grouped measurements in three levels of different importance.

Level 1 data are of highest priority to fulfill the scientific objectives of the GO-SHIP cruises. Level 1 data should be collected at least once per decade on all lines of the GO-SHIP network. This recognizes that lines occupied at higher frequencies (yearly, biennial) do not need to undertake all level 1 measurement on all re-occupations.

Level 1 data:

- Dissolved inorganic carbon (DIC)
- Total Alkalinity (TAlk)
- pH
  (Note: any two of the above)
- CTD pressure, temperature, salinity (calculated)
- CTD oxygen (sensor)
- Bottle salinity
- Nutrients by standard auto analyzer (NO_3/NO_2, PO_4, SiO_3)
- Dissolved oxygen
- Chlorofluorocarbons (CFC-11, -12, -113) and SF_6
- Surface underway system (T, S, pCO_2)
- ADCP shipboard
- ADCP lowered
- Underway navigation and bathymetry
- Meteorological data

Level 2 data are highly desirable as augmentation and addition for the science objectives executed on GO-SHIP cruises. GO-SHIP recommends that level 2 data should be collected when possible.

Level 2 data:

- Discrete pCO_2
- ^{14}C by AMS
- CCl_4
- $\partial^{13}C$ of DIC
- Dissolved organic carbon
- Dissolved organic nitrogen
- Fe/trace metals
- CTD Transmissometer
- Surface underway system (pCO_2, nutrients, O_2, Chl, skin temperature)

Level 3 data are ancillary measurements that often benefit from being taken in conjunction with the core measurements and/or address a scientific question unique to the region of investigation. They are collected according to opportunity and space available. They should not significantly interfere with sampling of level 1 or 2 parameters, and may be regional or specific to an individual cruise.

Level 3 data (examples):

- Chlorophyll
- Primary production
- HPLC pigments
- Experimental continuous analyzers (such as pH, DIC, and TAlk, and O_2/Ar)
- $\partial^{15}N$
- NO_3
- ^{32}Si
- $\partial^{18}O$ of H_2O
- NH_4
- Low level nutrients
- Total organic phosphorus
- Upper ocean optical
- Isotopes of O_2
- N_2, Ar, O_2
- Methyl halides
- DMS

Data Policy Timelines

All level 1, 2, and 3 data obtained on GO-SHIP cruises shall be openly accessible to the community at large. The following timelines for submission of data are in place. All data should be publically available preferably through a central data management site. CCHDO (cchdo.ucsd.edu) has agreed to serve all GO-SHIP data.

Within 5 weeks of the cruise, released to the relevant data management structure:

- Preliminary CTD (pressure, temperature, salinity, oxygen if measured)
- A merged bottle data file including preliminary discrete salinity, oxygen, nutrients (and inorganic carbon system components)
- Preliminary CFC-11, CFC-12, CFC-113, SF_6
- Underway data, including continuous (1-minute) navigation, bathymetry, shipboard meteorological measurements, temperature, salinity, pCO_2 (if measured).
- Shipboard ADCP data

Within 6 months of the cruise, presuming the 5-week releases of CTD and discrete salinity data:

- Final salinity, oxygen, nutrients, CFC, CTD data
- Final underway data
- Final shipboard ADCP data
- Final carbon system parameters (Total CO_2 and Total Alkalinity required; pH, pCO_2 if measured)
- CDOM if measured
- Lowered ADCP (if measured)
- Any other Level 2 measurements

Within 6 months of shore-based analysis:

- Tritium/helium
- ^{14}C and ^{13}C
- DON if measured

Within 2 years of analysis:

- Any other (level 3) observations. Those based on discrete bottle samples should be submitted to the hydrographic data management structure and merged with the other bottle data.
- Underway data should be submitted to the underway data management structure to be merged with the Level 1 and 2 underway data.
- Other discrete sampling programs that are likely to be carried out on many of the cruises, such as transmissometry and optics, should be submitted to the relevant data management groups (examples are a JGOFS SMP project for global transmissometry, and the NASA DAC for optics).